

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		Using 18 seconds $2.78 \times 10^{-2} / 2.8 \times 10^{-2} / 2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3$ marks $0.0277777777777778 / 0.028 / 0.02631579 / 0.03 = 2$ marks (not standard form) $2.7 \times 10^{-2} = 2$ marks (incorrectly rounded) $2.78 \times 10^{-1} / 2.63 \times 10^{-1} = 2$ marks (incorrect conversion) Using 19 seconds $2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3$ marks $0.027 = 1$ mark (not standard form and incorrect rounding) NOT taking volume into account Using 18 seconds $1.39 \times 10^{-2} / 1.4 \times 10^{-2} = 2$ marks $1.38 \times 10^{-2} = 1$ mark (incorrectly rounded) $0.0139 / 0.014 = 1$ mark (not taking standard form into account) $1.3 \times 10^{-2} = 1$ mark (incorrectly rounded) Using 19 seconds $1.32 \times 10^{-2} / 1.3 \times 10^{-2} = 2$ marks $0.0132 / 0.013 = 1$ mark (not taking standard form into account) $1.31 \times 10^{-2} = 1$ mark (incorrect rounding) OR Allow $2 \times 0.25 = 0.5$ for 1 mark		3		3	3	
		(ii)		Cooling in the colorimeter/ Accuracy of equipment used to measure volumes/ not all volumes transferred (1)			1	1		1